Sub Code:NEC303

Paper Id: 130316 Roll No:

dB TECH
(SEM-III) THEORY EXAMINATION 2019-20
SIGNAL & SYSTEMS

Time: 3 Hours ***Total Marks: 100***

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

a.	Find the fundamental period of the given signal. $X(n)= \sin(6\pi n/7 + 1)$.
b.	What is the relationship between z transform and Fourier transform?
c.	Find the Fourier transform of $x(t) = \sin(\omega t)\cos(\omega t)$.
d.	Examine whether the signal is periodic or not. If periodic then find out the period. $x(t)= \sin(10t+1) -2\cos(5t-2)$.
e.	Consider a discrete time system with input x(n) and output y(n):$y(n)=n[x(n)]^2$. Is this system time variant or time invariant?
f.	State convolution property of Z transforms.
g.	An LTI system is described by the differential equation $dy(t)/dt +4yt= x(t)$.Determine its impulse response h(t) and then H(f)?
h.	Find out the Laplace transform of the signal with its ROC. $X(t)= e^{-t}.u(t) + e^{-u}.u(t)$
i.	Prove the time differentiation property of Fourier transform
j.	Verify whether the given system described by the equation is linear and time invariant. $X(t)=t^2$

SECTION B

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2. Attempt any *three* of the following: 10x3=30

a.	Find the continuous time Fourier transform of the gate/ rectangular signal. Also plot its magnitude response.
b.	Define invertible system and state whether the following systems are invertible or not. 1) $Y(n)= x(n)$ 2) $Y(n)= x^2(n)+1$
c.	A discrete time system is given as $y(n)= y^2(n-1)+x(n)$. A bounded input of $x(n) =2n$ is applied to the system. Assume that the system is initially relaxed: check whether the system is stable or unstable.
d.	
e.	Given $x(t) = 5\cos$, $y(t)= 2e^{-t}$, find the convolution of x(t) and y(t) using Fourier transform.

SECTION C

3. Attempt any *one* part of the following: 10x1=10

a.	Determine the z-transform of following sequences with ROC i) U[n] ii) –u[-n-1] iii) $X[n]= a^nu[n] – b^nu[-n-1]$
b.	State and prove initial and final value theorem for z transform.

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4. Attempt any *one* part of the following: 10x1=10

a.	Explain sampling and quantization. What is the difference between uniform and non uniform sampling and quantization?
b.	i) Determine whether the following signal is energy or power signal. X(n) =u[n]- u[n-6]. ii) sketch the following signal y(t)= π((t/3)-2) + π(2t-3.5)

5. Attempt any *one* part of the following: 10x1=10

a.	i) Determine whether the system is BIBO stable or not. Y(n)= max[x(n+1), x(n), x(n-1)] ii) Check whether the system is static / dynamic and causal/ non causal and why? Y(n) = $\log_{10}\|x(n)\|$
b.	Find the Laplace transform for the parabolic function x(t)= t^2. e^{-3t}. u(t)

6. Attempt any *one* part of the following: 10x1=10

a.	Determine the total response of the difference equation Y(n)+4y(n-1)+4y(n-2)= $(-2)^n$u(n) Where y(-1)=0 and y(-2) = 1
b.	i) Using Fourier transform, find the convolution of: $X_1(t)= e^{-2t}.u(t)$ $X_2(t)= e^{-3t}.u(t)$ ii) calculate the DTFT of the following using properties of DTFT x(n)= u(n+3) – u(n-3)

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7. Attempt any *one* part of the following: 10x1=10

a.	Determine the total response of the differential equation $d^2y(t)/dt^2+3dy(t)/dt+2y(t)= x(t)$ Where y(0)= 3, y(0)=4, x(t)= $4e^{-2t}$ and t≥0.
b.	Compare between CTFT and DTFT with examples.

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